TerraAlert AI

Present by Team-10

MEET OUR TEAM MEMBERS

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THE REAL-WORLD PROBLEM

Why Is This Project Needed?

- Landslides, floods, and droughts can affect communities, farms, roads, and buildings.
- Early warning information may not be easily available in every area.
- People may notice soil changes but may not understand the possible risks.
- Professional monitoring equipment can be expensive or unavailable.
- Delayed risk awareness can increase damage and reduce preparation time.

OUR SOLUTION

What if one soil photo could provide an early warning?

TerraAlert AI combines:

- AI soil-image classification
- Rainfall, slope and nearby water
- Ground cracks and vegetation conditions

It turns these environmental signs into clear Landslide, Flood and Drought risk levels within seconds.

- One Soil Image • Multiple Risk Factors • Three Risk Assessments

HOW THE SYSTEM WORKS

1. Upload a soil image.
2. The system identifies the visible soil condition.
3. Enter environmental information:
   - Rainfall level
   - Land slope
   - Nearby water
   - Visible ground cracks
   - Vegetation condition

HOW THE SYSTEM WORKS(con’t)

4.The system combines all the factors.

5.It calculates preliminary risks for:

- Landslide
- Local Flood
- Drought

6.Results are displayed with scores and risk levels.

MAIN FEATURES

Key System Features

- Soil image upload
- Dry, Normal, and Wet soil classification
- AI Model Prediction Mode
- Demonstration Mode for reliable presentation
- Confidence score for AI predictions
- Landslide, flood, and drought risk scores
- Low, Medium, and High risk levels
- Assessment history
- Dashboard charts and saved records
- Local laptop operation without paid APIs

BENEFITS AND ADVANTAGES

Why Is TerraAlert AI Useful?

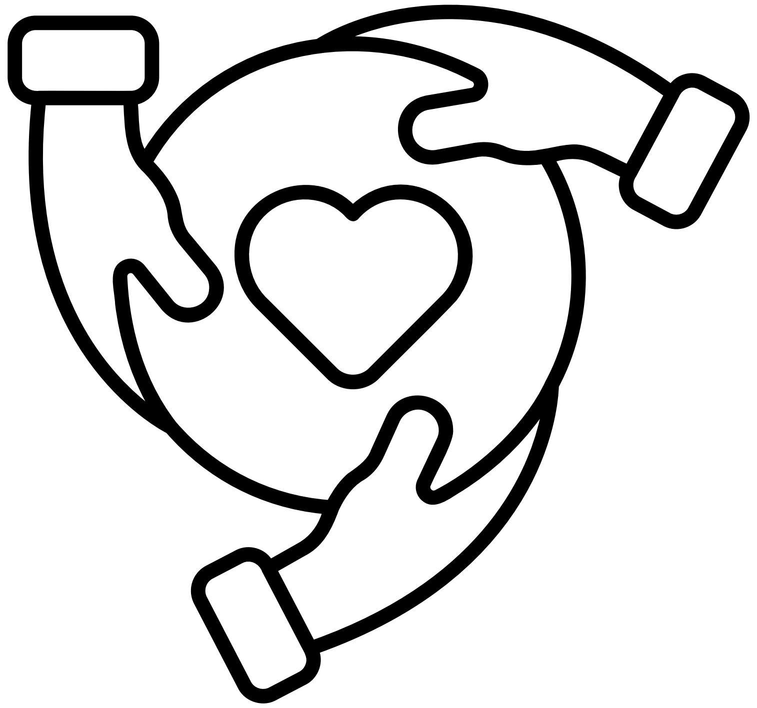
- Early Awareness: Highlights possible risks before conditions become more serious.
- Multiple-Risk Analysis: Checks three environmental risks in one system.
- Better Decisions: Combines several warning signs instead of using only one factor.
- Easy to Use: Users only need a soil image and simple environmental information.
- Low-Cost Prototype: Runs locally on a laptop without external hardware.
- Clear Results: Uses simple scores, charts, and risk levels.
- Record Keeping: Saves previous assessments for comparison and monitoring.
- Expandable: Can be improved with sensors and live environmental data.

Future Improvements

- Connect soil-moisture and rainfall sensors.
- Use live weather and rainfall data.
- Add GPS, elevation, and terrain information.
- Train the AI model with a larger local dataset.
- Add map-based risk areas and mobile alerts.
- Test the system with environmental experts.

CONCLUSION

TerraAlert AI combines soil-image analysis and environmental information to support early risk awareness and better preparation.Smart Observation. Early Awareness. Safer Decisions.



THANK YOU FOR YOUR ATTENTION!